Trainings

General Information

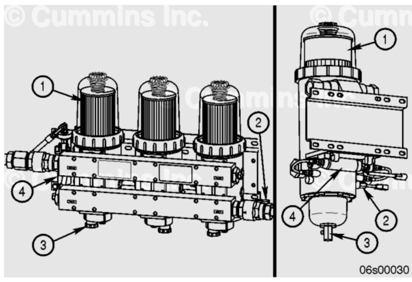
Note : This procedure applies to Stage 1 fuel filters on engines with electronically actuated injectors only.

- Use the following procedure for mechanically actuated injectors. Refer to Procedure 006-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/28/28-006-015-om.html)
- Use the following procedure for stage 2 fuel filters. Refer to Procedure 006-076 in Section 4. (/qs3/pubsys2/xml/en/procedures/28/28-006-076-om.html)

with Electronically Actuated Injector

Industrial and Power Generation Applications

- The Stage 1 filter mounting location is original equipment manufacturer (OEM)-dependent, provided the Cummins Inc. applications guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- Industrial and Power Generation Applications are explained in the following steps.



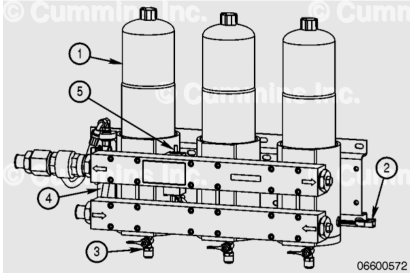
The Stage 1 filters are mounted off the engine. The fuel priming pump is mounted with the Stage 1 filter assembly.

The FH234 series is on the left of the illustration and the FH239 series is on the right. The filter consists of the following components:

1. Stage 1 filter
2. Water-in-fuel sensor
3. Drain valve
4. Lift pump.

Note : The following steps show the FH234 series, but the FH239 series steps are very similar.

Note : The filters shown throughout this procedure may not be the same on all engines. Although different in appearance, the procedure remains the same.



Marine Applications without individual filter shutoff valves

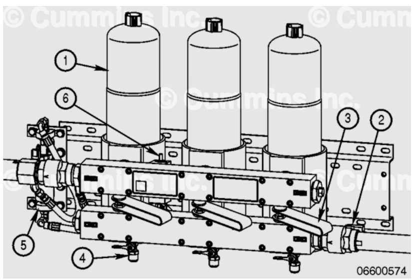
- The Stage 1 filter mounting location is OEM-dependent, provided the Cummins Inc. application guidelines are met. Contact a Cummins® Authorized Repair Location for further information.
- The Marine application engines are equipped with two different Stage 1 filter systems. They are identified as Triplex or Quadplex (filters with individual shutoff valves) and Dual or Triple (filters without individual shutoff valves). Triplex and Quadplex filter applications enable the engine to remain in operation while the filters are changed. Dual or Triple filter applications require the engine to be shut down for service.
- The maintenance interval can **not** be exceeded under any condition, regardless of the Stage 0 design.
- The pop-up indicator on the Sea Pro™ Stage 1 filter is used to indicate when the filters are contaminated before the standard interval. It is **not** to be used as an extended life monitor.
- When the indicator pops up, all filters **must** be changed.
- Dual or Triple filter applications are explained in the following steps.

The Dual or Triple Sea Pro™ filter is equipped with an electric priming pump mounted to the filter bracket.

The Sea Pro™ Dual or Triple filter consists of the following:

1. Stage 1 filter
2. Water-in-fuel sensor
3. Drain valve
4. Electric priming pump
5. Differential pressure “pop-up” indicator.

Note : The filters shown throughout this procedure may not be the same on all engines. Although different in appearance, the procedure remains the same.



Marine Triplex/Quadplex Filter Applications

- The Stage 1 filter mounting location is off the engine and therefore OEM-dependent. Cummins Inc. application

guidelines **must** be met. Contact a Cummins® Authorized Repair Location for further information.
- This procedure explains Stage 1 fuel system filter changes during engine operation. If the engine uses a Triplex/Quadplex filter, and the intent is to change filters with the engine shut off, follow the Dual/Triple filter procedures.
- Triplex/Quadplex applications are explained in the following steps.

- The maintenance interval can **not** be exceeded under any condition, regardless of the design.
- The pop-up indicator on the Sea Pro™ Stage 1 filter is used to indicate when the filters are contaminated before the standard interval. It is **not** to be used as an extended life monitor.
- When the indicator pops up, all filters **must** be changed.

The valves are located between the inlet and outlet fuel manifolds on the filter assembly.

The Triplex filter consists of the following:

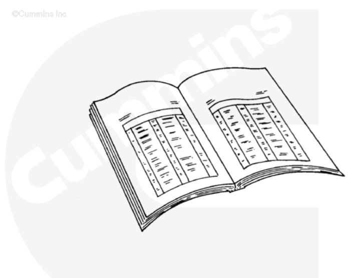
1. Stage 1 filters
2. Water-in-fuel sensor
3. Shutoff valves
4. Drain valves
5. Electric priming pump
6. Differential pressure “pop-up” indicator.

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



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Industrial and Power Generation Applications

- The Stage 1 filtration for Industrial and Power Generation Application QSK38/50 MCRS engines consists of a Dual/Triple Industrial Pro™ filter head. The Stage 1 filtration **must** be used during normal engine operation.

Shut down the engine.

Close the fuel supply shutoff valve.

⚠ WARNING ⚠

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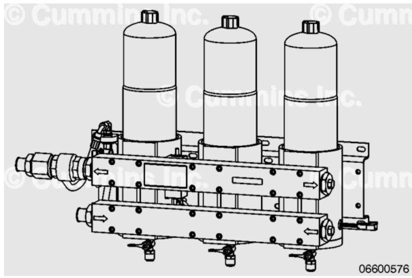
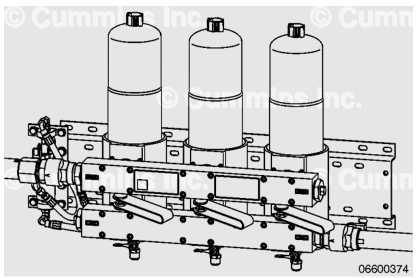


Marine Dual/Triple filter Applications

- The Stage 1 filtration for the Dual/Triple filter Marine Application consists of two or three Sea Pro™ filter heads. QSK38 engines have a dual filter head while QSK50 engines have a triple filter head. The Stage 1 filtration **must** be used during engine operation.

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Note : This procedure explains Triplex fuel system filter changes with the engine operating. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shutoff, follow the Dual/Triple filter procedures.

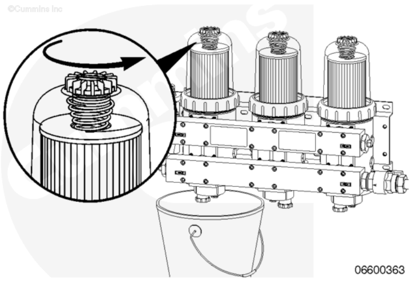
Marine Triplex/Quadplex Applications

- The Stage 1 filtration for the marine engine consists of a Triplex Sea Pro™ 5 filter head on QSK38 engines and a Quadplex filter head on QSK50 engines. **Only** one filter element at a time can be isolated or removed from the fuel flow during engine operation.

With the engine operating, move the valve handle to the OFF position on the Stage 1 filter to be changed.

Remove

with Electronically Actuated Injector



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

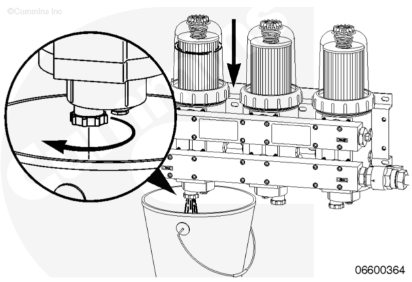
Note : Industrial and Power Generation applications are explained in the following steps.

Industrial and Power Generation Applications

- The fuel **must** be drained from the filter head prior to removing the filter element.
- Do **not** allow fuel to drain onto the ground. Drained fuel **must** be collected and disposed of in accordance with local environmental regulations.

Place a suitable container under the Stage 1 fuel filter to be replaced.

Use the collar/vent cap wrench, Part Number 3944451S, to open the vent cap to break the air lock in the filter.

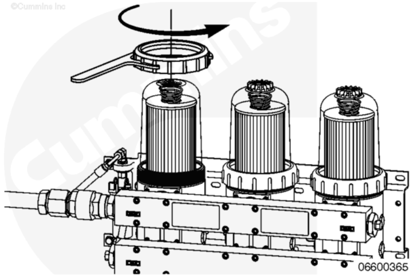


Open the fuel drain valve and allow the fuel level to drain to a point below the collar.

Close the drain valve.

Use the collar/vent cap wrench, Part Number 3944451S, to loosen the collar.

Remove the collar.

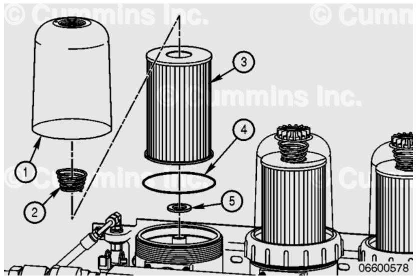


Remove the clear cover (1), filter spring (2), fuel filter element (3), and o-ring (4).

Remove the sealing grommet (5) from the filter stud.

Discard the o-ring (4) and sealing grommet (5).

Note : The FH239 series has two filter elements that are stacked together. Make sure both elements are removed and replaced at the same time.

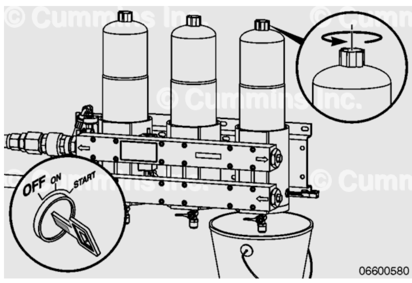


⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

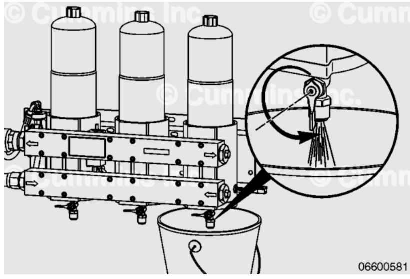


Marine Dual/Triple Filter Applications

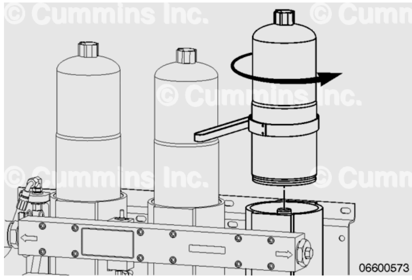
- The keyswitch **must** remain in the OFF position while draining the Stage 1 filter. This will prevent excessive air intake into the fuel system.
- Place a suitable container under the Stage 1 fuel filter to be replaced.
- Open the vent cap to break the air lock in the filter.

Open the fuel drain valve and allow the filter to drain.

Close the drain valve when the drainage slows to a trickle.



Remove the Stage 1 fuel filter with filter wrench, Part Number 3400157, or equivalent.



⚠ WARNING ⚠

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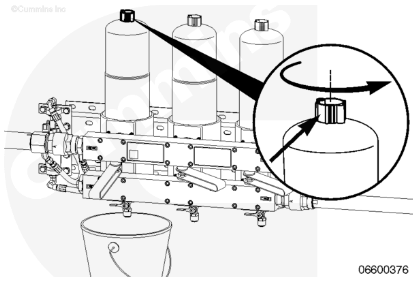
⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.



Note : This section explains Triplex/Quadplex fuel system filter changes with the engine running. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shut off, follow the Dual/Triple filter procedures.

Marine Triplex/Quadplex Applications

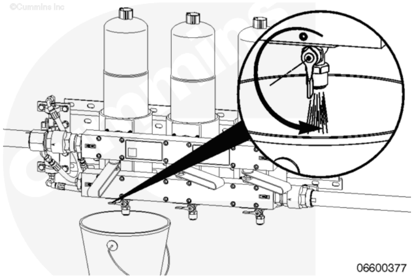
- After the filter element has been removed, the selector valve **must** remain in the OFF position.

Place a suitable container under the Stage 1 fuel filter to be replaced.

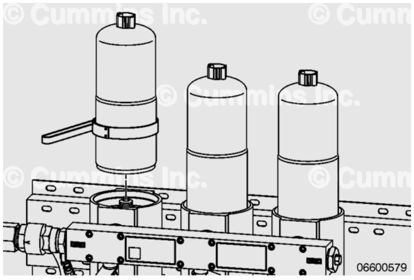
Open the vent cap to break the air lock in the filter.

Open the fuel drain valve and allow the filter to drain.

Close the drain valve when the drainage slows to a trickle.



Remove the fuel filter. Use filter wrench, Part Number 3400157, or equivalent.

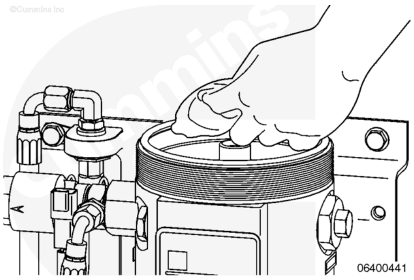


Clean

with Electronically Actuated Injector

Note : The Industrial application filter is shown throughout this procedure. Although different in appearance, the procedure remains the same for Industrial and Marine applications.

Check the sealing surface of the filter body. Wipe the surface with a clean, lint-free cloth to remove any debris or dirt.



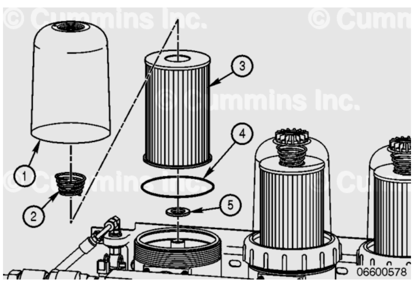
Install

with Electronically Actuated Injector

Note : Verify the used o-ring has been removed and discarded before installing the new o-ring.

Industrial and Power Generation Applications

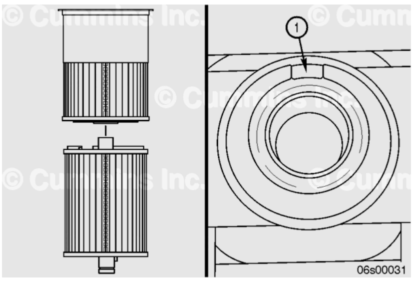
- Stage 1 filter element replacement includes the appropriate o-ring and sealing grommet.



- The o-ring and grommet **must** be replaced with the filter element to make sure of proper operation.

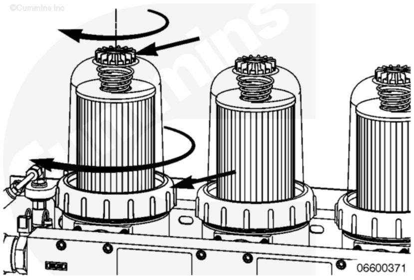
On the Stage 1 filter, for the Industrial and Power Generation engines, install a new o-ring (4), filter element (3) (supplied with a sealing grommet (5) inserted into the filter element), filter spring (2), and clear cover (1).

The FH239 series has two filter elements that are stacked together. The bottom element has a key that **must** match the keyway (1) in the filter head.



Install the vent cap and collar onto the clear cover and hand tighten **only**.

Do **not** use tools to tighten the collar.



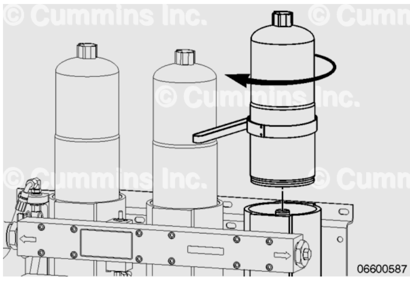
Marine Dual/Triple filter Applications

- The Stage 1 filter element replacement includes the appropriate o-ring.
- The o-ring **must** be replaced with the filter element to make sure of proper operation.

Install the Stage 1 filter on the filter head.

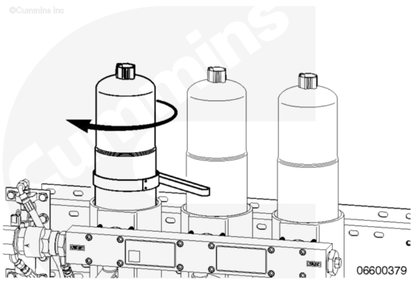
Turn the filter until the o-ring touches the surface of the filter head.

Tighten the filter an additional ½ to ¾ of a turn.



Note : This procedure explains Triplex/Quadplex fuel system filter changes with the engine operating at idle. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shut off, follow the Single filter procedures.

Marine Triplex/Quadplex Applications



- The Stage 1 filter element replacement includes the appropriate o-ring.
- The o-ring **must** be replaced with the filter element to make sure of proper operation.

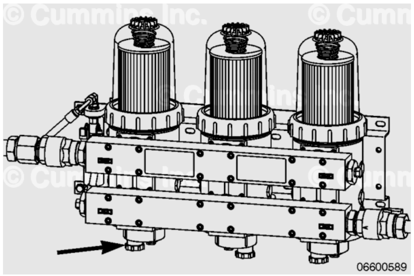
Install the Stage 1 filter on the filter head. Turn the filter until the o-ring touches the surface of the filter head.

Tighten the filter an additional ½ to ¾ of a turn.

Prime

with Electronically Actuated Injector

Industrial and Power Generation Applications

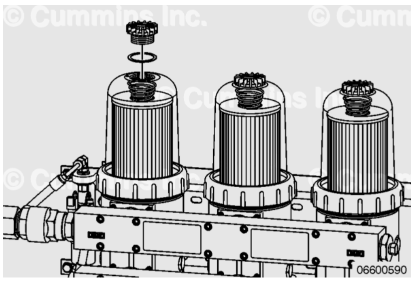


- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- The procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows:

Verify the drain valve on the Stage 1 filter is closed at the base of the filter.

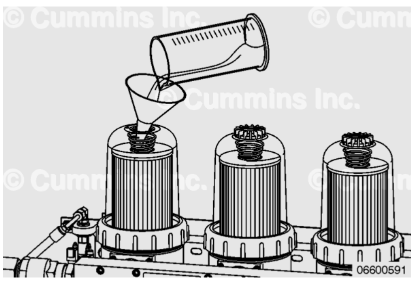
Close the fuel supply shutoff valve.

Remove the vent cap from the top of the filter.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



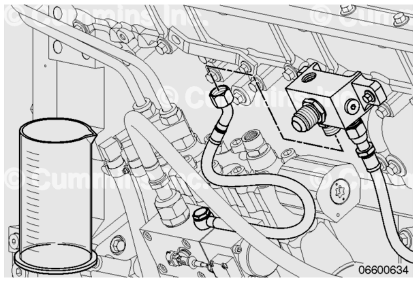
Fill the filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel supply shutoff valve.

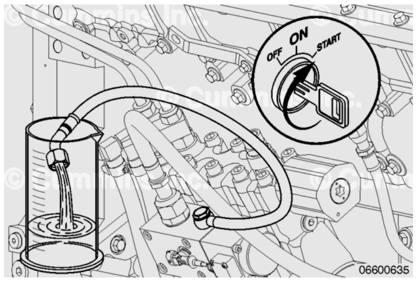
Remove the air bleed line from the drain manifold block and place into a collection container.



Turn the keyswitch ON and observe fuel exiting the air bleed line.

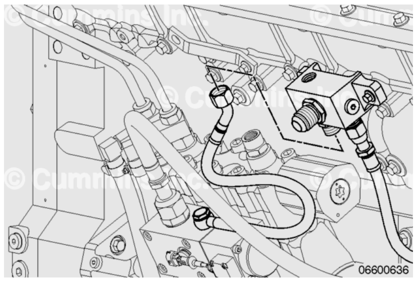
When a steady stream of fuel is observed, turn the keyswitch OFF.

If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON. Repeat as necessary to observe a steady stream of fuel.



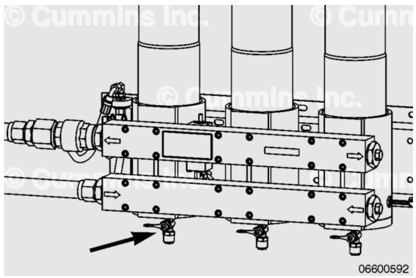
Install the air bleed line to drain the manifold block.

Torque Value: 45 n•m [33 ft-lb]



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



Marine Dual/Triple Filter Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.

- The procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Stage 1 Priming:

Verify the drain valve is closed at the base of the Stage 1 filter.

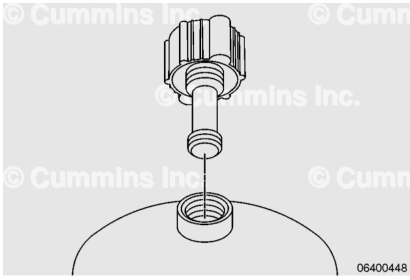
Close the fuel supply shutoff valve.

Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

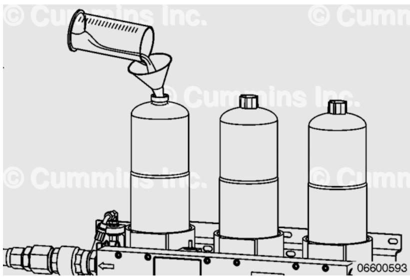
Unscrew the retainer shaft from the threaded boss in the top of the filter can.

Remove the vent cap from the top of the can.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



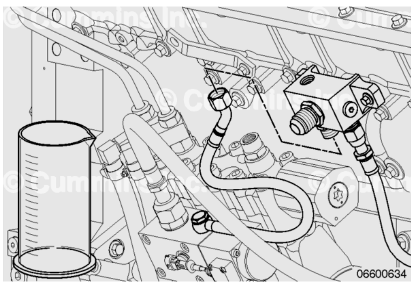
Fill the filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel supply shutoff valve.

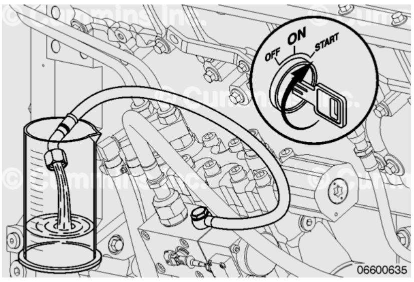
Remove the air bleed line from the drain manifold block and place it into a collection container.



Turn the keyswitch ON and observe fuel exiting the air bleed line.

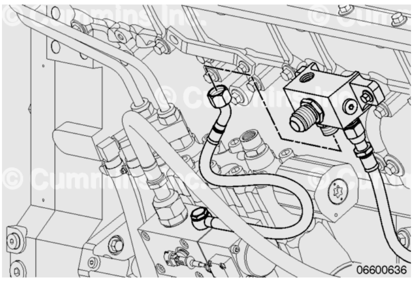
When a steady stream of fuel is observed, turn the keyswitch OFF.

If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON. Repeat as necessary to observe a steady stream of fuel.



Install the air bleed line to drain the manifold block.

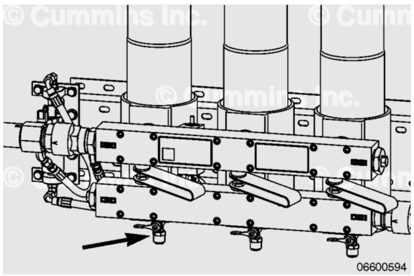
Torque Value: 45 n•m [33 ft-lb]



Note : This procedure explains Triplex/Quadplex fuel system filter changes with the engine running at idle. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shut off, follow the Dual/Triple filter procedures.

Marine Triplex/Quadplex Applications

- Stage 1 filters **must** be primed prior to resuming filter operation.
- The procedure for priming stage 1 filtration is as follows:

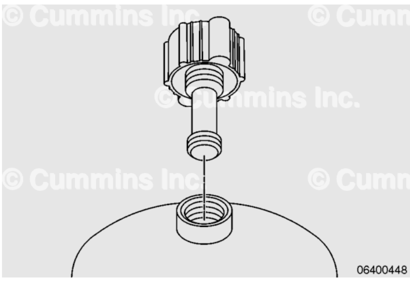


Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

Unscrew the retainer shaft from the threaded boss in the top of the filter can.

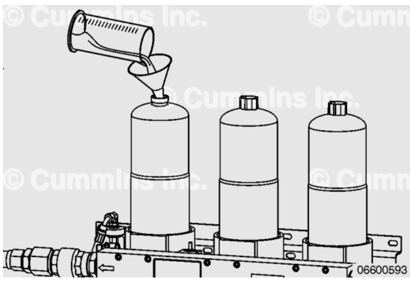
Remove the vent cap from the top of the can.



Fill the filter with clean diesel fuel.

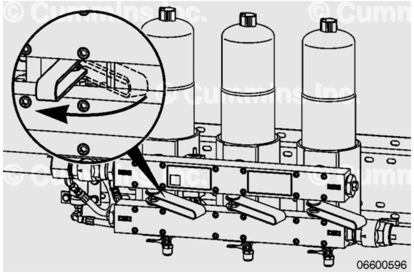
Install the vent cap.

Tighten the vent cap by hand **only**.



Slowly open the valve handle to the ON position, allowing excess time to purge air from the system.

Repeat the procedure for the remaining filter replacement.

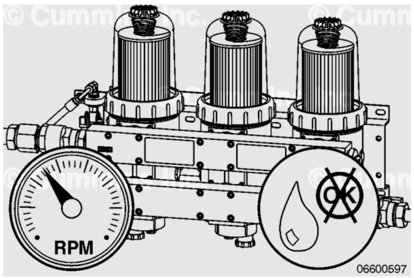


Finishing Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

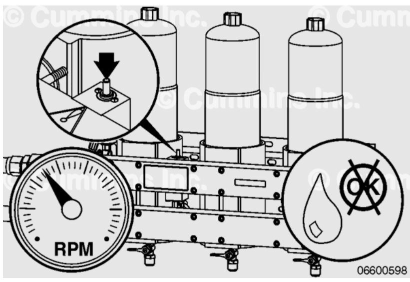


Industrial and Power Generation Applications

- The Industrial Pro™ filter **must** be checked for leaks.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Operate the engine for 1 to 2 minutes and check for leaks.

Marine Dual/Triple Filter Application

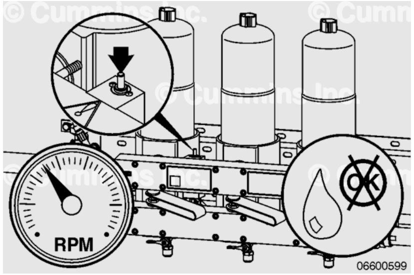
- The Sea Pro™ filter head **must** be checked for leaks.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Reset the Delta P indicator by firmly pressing down on the pop-up indicator.
- Operate the engine for 1 to 2 minutes and check for leaks.



Marine Triplex/Quadplex Application

- Check the Sea Pro™ filter head for leaks while the engine is running at idle.

- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Reset the Delta P indicator by firmly pressing down on the pop-up indicator.
- Operate the engine for 1 to 2 minutes above idle or reduced capability and check for leaks.



Last Modified: 13-Aug-2013